Introduction
Ensuring hospitals are clean and safe is an essential component in the provision of effective healthcare. A clean and tidy environment is an outward manifestation of the good patient care practice. It is fundamental in assisting patients to recover and help in the prevention and/or control of the spread of healthcare associated infections.

Purpose
This policy provides sets of standards and practices in cleanliness of the hospital environment to protect the patient/resident, staff and visitors in the healthcare facility by minimizing the possible spread of infections. For the techniques and procedure of cleaning in hospital please refer to Royal Hospital cleaning standard operating procedures.

Scope
This policy must be followed by housekeeping staff, nurses, health & safety officer, infection prevention & control staff, engineering staff.
This policy must be read in accordance with other polices as follow:
- Terminal cleaning policy IPC-SOP-1-Vers.1.0
- Laundry and Linen Service policy DGA-POL-7-Vers.01
- Management of clinical waste policy HSE-POL-4-Vers.2.0
- Cleaning & disinfection of non-critical medical equipment policy POL/IPC/24/Ver.3
- Hydrogen peroxide policy IPC-POL-54-Vers.2.0
- Disinfection Wipes utilization policy IPC-POL-64-Vers.1.3

Definitions
- **Cleaning:** The physical removal of foreign material (e.g., dust, soil) and organic material (e.g., blood, secretions, excretions, microorganisms). Cleaning physically removes rather than kills microorganisms. It is accomplished with water, detergents and mechanical action.
- **Environmental Cleaning:** The removal of all dirt and soiling to ensure the entire area is hygienically cleaned at all times.
- **Basic cleaning:**It is done daily. It includes dust and dirt removal, waste disposal and cleaning of windows and surfaces. Table 1 is list component of basic routine cleaning.
- **Hospital Grade cleaning:**It includes all cleaning components of basic cleaning and the following:
  - The high touch surfaces in patient's areas are disinfected after cleaning with hospital grade disinfectant.
  - Non-critical medical equipment are cleaned and disinfected between clients/patients.
  - Cleaning practices are periodically monitored and audited.
- **Deep Cleaning:** It is the periodic cleaning of wall and ceiling surfaces in laboratories, theatres, specialist areas, wards and kitchens excluding the routine day to day cleaning of sanitary fittings and floors, and spot cleaning of walls and doors in such areas.
- **Terminal cleaning:** It is a thorough cleaning of a patient room following discharge in order to remove germs that might be transferred to the next patient in the room. Refer to terminal cleaning procedure

- **Clinical Areas:** The areas used to deliver clinical care to patients where the need for high standards of hygiene is paramount on a day-to-day basis. This includes client/patient/resident units (including nursing stations), procedure rooms, bathrooms, clinic rooms, and diagnostic and treatment areas.
- **Non-Clinical areas:** The areas not included under the category of Clinical Areas such as administration area, medical records, public areas such as lobbies and waiting rooms, offices, corridors, elevators, stairwells and service areas. These should have different cleaning regimen than clinical areas.
- **Disinfection:** The destruction of pathogenic and other kinds of micro-organisms by thermal or chemical means.
- **Disinfectant:** A chemical that rapidly kills or inactivates most infectious agents. Disinfectants are only to be used to disinfect and must not be used as general cleaning agents, unless combined with a cleaning agent as a cleaner-disinfector. Skin antiseptics must never be used as environmental disinfectants (e.g., alcohol-based-hand rub, chlorhexidinegluconate).
- **Disinfectant Wipes:** Ready-to-use wipes for point-of-care cleaning and disinfecting of non-critical patient equipment. For more information refer to Disinfection wipes utilization policy
- **Outbreak:** Where the incidence of infections or colonization is greater than the expected rate within a specific area over a defined period of time. An enhanced cleaning is required to control the spread of micro-organisms during outbreak which environmental departments should allow for surge capacity (i.e., additional staff, supervision, supplies, equipment) during outbreaks as determined by the outbreak management committee.
- **Functional Areas:** A functional area refers to any area in a healthcare facility that requires cleaning. The functional areas have been grouped under four risk categories: extreme, high, medium and low. Cleaning & disinfection of these areas planned scheduled as per above mentioned risk. Refer to Table 2 for classification of each hospital area.
- **High-Touch Surfaces:** High-touch surfaces are those that have frequent contact with hands. Examples include doorknobs, telephone, call bells, Bedside table, Chair, bedrails, light switches, and Toilet seat, wall areas around the toilet and edges of privacy curtains. Surfaces closer to the patient/resident pose a greater risk for transmission than those situated further away. Furthermore, frequently touched surfaces are more likely to harbour and transmit microbial pathogens. It would therefore, be cost-effective to concentrate cleaning resources on high risk, high touch surfaces.
- **Low-Touch Surfaces:** Surfaces that have minimal contact with hands. Examples include walls, ceilings, mirrors and window sills.
- **Damp Dusting:** Use of a clean, low-linting cloth moistened with disinfectant for cleaning.
- **Microfiber:** Are densely constructed polyester and polyamide (nylon) fibers that are approximately 1/16 the thickness of a human hair. The positively charged microfibers attract dust and bacteria (which have a negative charge), using a combination of static attraction and capillary action, from the surface pores of most surface and flooring materials and hold it tightly so that it is not redistributed around the room during cleaning. Microfiber materials are more absorbent than conventional cloths or cotton-loop mops, enabling them to hold six times their weight in water. Some of these materials can be wet with disinfectants.

Roles & Responsibilities

- **HOD of environmental Service:**
    - Maintaining the cleaning contract and hospital policy and procedures
    - Ensuring that the education and training activities are carried out for all and new cleaning staff before assignment and on a recurring basis
    - Supervising the cleaning audit process and utilizing the results for improvement and giving feedback to stakeholders.
    - Ensuring that cleaning supplies and equipment are available in required quantities and in good condition (i.e., preventing stock-outs/ non interruption)
    - Addressing stakeholders concerns and patient comments about the cleaning process
    - Communicating with the cleaning company on any of the cleaning and disinfection issues
    - Coordinating with the cleaning company to develop schedules of cleaning activities in all areas

- **Director of Nursing:**
    - Ensure that the nursing practice is in line with this policy and monitor the implementation of the standards listed in here.

- **Ward Nurse in charge:**
    - Charge Nurses are responsible for ensuring safe working conditions within their clinical area. This includes all aspects of environmental cleanliness.
    - Charge Nurses have the authority to require local cleaning services to act on any problems identified.

- **Cleaning supervisor:**
    - Ensure staff training and compliance with cleaning tasks procedures, safety and compliance with PPE.
    - Audit the compliance and institute actions whenever corrective measures required.

- **Infection Prevention & Control Team:**
    - Support, advice and guidance on specific and/or specialist cleaning requirements
    - Monitor compliance to standards and practices stated in this policy

- **All HCW:**
    - Ensuring standards of cleanliness are maintained to protect patients and others.
    - Understanding their individual responsibilities in relation to environmental cleaning and decontamination of equipment.
    - Understanding the process for raising concerns in relation to failures in cleaning processes.

Health & Safety:

All staff involved in the cleaning procedure must follow the following:

A. Personal safety:

Every employee must accept responsibility for his or her personal safety. Failure to comply with instructions, with written procedures, or with your training will increase the risk of you causing an accident which harms yourself or someone else. You have a responsibility to co-operate with your employer by working safely and efficiently.

Always ensure that you:

- Take reasonable precautions to safeguard the health, safety and welfare of yourself and others who may be affected by your work
- Observe all health and safety rules and procedures laid down by your employer
- Always use any health and safety equipment that your employer provides you with
- Do not intentionally or recklessly interfere with or misuse anything that is provided in the interest of health and safety
- Report any faulty equipment to your supervisor without delay
- Alert your supervisor without delay if you notice a potential hazard
- Report all incidents which may lead to injury, completing an accident form in line with your employer's policy
- Do not attempt any task for which you do not have properly signed training.

B. General Health and Safety rules:

- Work in a controlled and systematic way, do not rush or allow yourself to become distracted
- Walk between tasks, do not run
- Use only authorized access and exit routes into buildings wards and departments
- Always lift and move items in accordance with your manual handling training
- Keep work areas tidy and tidy up as you work.

C. Using electrical equipment:

- Always follow the manufacture equipment manual on cleaning, handling and storage of equipments.
- Check equipment before and after use for damage or wear which might potentially be dangerous, including breaks or cracks on the plug or cable. Any damaged or untested equipment should be reported immediately to a supervisor, who will ensure that the equipment is labeled as unfit for purpose and removed from use.
- When using electrical equipment, plan your work to ensure that the cable will always be behind the machine. Ensure that the switch is in the "off" position before plugging machinery into a wall socket. Always unplug machinery before changing fittings or settings.
- Ensure hands are dry when plugging or unplugging electrical machinery.
- Always make sure to plug the machine into a free plug socket. If none are available, ask the person in charge before unplugging any other machinery.
- The electrical cable must never be draped over your shoulder or over the handle of the machine.
- Do not adjust or change the fittings on the machine when it is plugged in.

- Never use a piece of electrical equipment unless you have signed training in its use.
- Never leave unattended machinery attached to a power source.
- Never leave machinery unattended where it could be a source of danger to others.
- Should the machine switch off automatically, stop work and inform your supervisor.
- Water or steam should not be used directly on electrical sockets.

D. Guidance related to cleaning activities:

- Refer to the work schedule and identify the task to be performed.
- Identify the type of area in which cleaning is to be performed and select the correct colour-coded equipment for the task as mentioned below.
- Prepare cleaning solutions in strict accordance with the manufacturer's instructions and with your training.
- Do not mix chemicals and only use a cleaning product provided by your employer.
- Plan your work and where necessary temporarily move any items that might obstruct you to a new, safe location.
- Use hazard signs to warn other users of the area in which you are carrying out a cleaning task.
- After use, all equipment should be left clean, dry and tidy in a secure storage area, segregated according to colour-coding where appropriate.
- Determine whether personal protective equipment should be worn, or other additional precautions are necessary.

E. Clothing, jewelry and personal hygiene:

- All are obliged to adhere to hospital uniform code.
- Loose fitting clothes and jewelry may become caught in machinery and therefore must be avoided. Depending upon the nature of the task, a plain wedding band may be worn.
- If you wish to wear jewelry for religious, health or other reasons, you should seek the advice of your supervisor before commencing work. Long hair should be tied up off the collar.
- Good personal hygiene plays an important role in health and safety at work.

F. Safe manual handling

Training should be given in the correct method of lifting, carrying and pushing and pulling. If you develop a medical problem that would place you at risk when carrying out manual handling, you should alert your supervisor who should carry out a risk assessment. It is advisable to alert your supervisor if you become pregnant so that a personal risk assessment can be carried out.

G. Hand Hygiene for cleaning staff in healthcare environment:

- Hand must be washed before and after work is very important to stop transmitting germs.
- The staff must follow the 5 moments of hand hygiene when cleaning immediate patient environment. Alcohol hand rub is enough for this purpose.
- Hand hygiene after glove removal is essential.
- In the following situation require hand washing with soap and water:
  1. When hands are visibly dirty.

2. When hand become contaminated with body fluids
3. Cleaning an area where a patient has diarrhea and or vomiting. This is because alcohol hand rub does not kill some of the germs that cause diarrhea and vomiting

H. Cough Etiquette:

- A series of actions to taken when coughing or sneezing; these actions are designed to reduce the spread of respiratory illness to others.
- Cough etiquette should always be applied as a standard infection control precaution. Covering sneezes and coughs limits infected persons from dispersing secretions into the air.
- Helpful tips on how to perform cough etiquette:
    - cover the nose and mouth with disposable tissue when coughing, or sneezing
    - dispose of tissues in the nearest waste receptacle or bin after use
    - if no tissues are available, cough or sneeze into the inner elbow rather than the hand
    - if you accidentally cough or sneeze into your hand, wash them or use (Alcohol Based Hand Rub) ABHR as soon as practicable
    - Remember to turn away from other people when you cough or sneeze.

I. Personal protective equipment for cleaning staff:

Gloves:

- Prolonged wearing of gloves is not recommended both because of increased risk of irritant dermatitis from sweat and moisture within the gloves as well as breakdown of the glove material itself and risk tears.
- Inappropriate use of gloves, such as going from room to room in care areas with the same pair of gloves can facilitate the spread of micro-organisms.
- Glove usage should therefore be limited to when there is a risk of contact with blood, body fluids, secretions, excretions or equipment that may be contaminated with these fluids and for protection against harsh chemical fluids. Examples include:
    - Cleaning toilets and bathrooms
    - Cleaning up spillages such as blood, vomit and faeces
    - When cleaning environments which require transmission-based precautions.
    - Using harsh chemical fluids. Nitrile gloves are recommended for providing protection against chemicals.
- Gloves must be removed immediately after the activity for which they are used.
- Plastic gloves may be worn for cleaning tasks. These should be sturdy, suitable for purpose. Gloves should be inspected before use to ensure that they are intact.
- Reusable gloves (heavy duty) should be cleaned regularly between cleaning tasks. Use of gloves does not reduce the requirement for hand washing.
- Disposable vinyl gloves may be used for routine daily cleaning and disinfection procedure in client/patients' rooms not under isolation and in public areas and offices
- Latex free gloves should be available to the above specification where a latex allergy has been identified.
- Choosing the right type of gloves for cleaning duties it is important to assess and select the most appropriate glove for the activity about to be performed. Selection of gloves should be

based on a risk analysis of the type of setting, the task that is to be performed, likelihood of exposure to body fluids, length of use and amount of stress on the glove.

Apron:

Disposable plastic aprons should be worn for all cleaning tasks in which splashes to clothing are likely to occur and when there is a risk of contamination with blood, body fluids, secretions or excretions.

If a gown or apron is required to be donned, then it should be fluid-resistant, and it must be changed between patients, rooms and tasks.

For isolation areas:

For isolation rooms/bays/wards PPE to be worn: disposable isolation gown and disposable gloves and surgical or N95 mask according to the type of isolation.

J. Accidental exposure to blood or body fluids:

Housekeeping staff must report any inoculation injuries such as needle stick, other sharps injuries, bites, scratches and splash contamination of broken skin after immediate action been taken to their immediate supervisor as per management of needle-stick injuries policy **IPC-POL-45-Vers.1.0**.The staff must attend the emergency department for risk assessment and immediate management.

K. Waste management and spills cleaning:

Follow instruction outlined in management of clinical waste policy HSE-POL-4-Vers.2.0

L. Immunization of housekeeping staff:

The hospital immunization of healthcare workers policy IPC-POL-6-Vers.1.1must be followed.

M. Illness Reporting:

The staff should report illness such cough and fever, rashes, diarrhea, and vomiting. They should be evaluated for suitability of work.

General Cleaning Practices and Standards:

- Routine cleaning is necessary to maintain a standard of cleanliness.
- All chemicals should be properly labeled and stored to eliminate any potential risk of contamination and injuries.
- Adequate resources including human resources must be devoted to environmental Services in the hospital to ensure compliance with practices are accomplished and maintained
- The frequency of cleaning and disinfecting individual items or surfaces in a particular area or department depends on:
  - Whether surfaces are high-touch or low-touch
  - The type of activity taking place in the area and the risk of infection associated with it (e.g., critical care areas vs. meeting room)
  - The vulnerability of clients/patients/residents housed in the area

- The probability of contamination based on the amount of body fluid contaminating surfaces in the area. Refer to functional areas category risk (Appendix B and C)
- The cleaning staff shall allow adequate cleaning chemical contact time, as per the manufacturer's recommendation, after the application of the cleaning chemical onto the surfaces.
- The cleaning is performed from the least contaminated to the most contaminated item and the cleaning of isolation rooms is performed after completion of cleaning all the non-isolation rooms. This means that areas/elements which are low touch or lightly soiled should be cleaned before areas/elements which are considered high touch or heavily soiled. For example:
  - When cleaning a bathroom, the toilet should be cleaned last as it is likely to be the most contaminated element in that area
  - In a patient room, items that would be considered high touch would include the patient bed, call-bell, locker, overwaytable; light switches, control knobs, hand basin etc., and low touch areas would include the walls, windows and floors.

- The flow of cleaning should generally be from high to low reach surfaces. For example; when dusting horizontal surfaces in a patient room, high areas such as those above shoulder height should be done first followed by all other elements. Dusting technique should not disperse the dust, (i.e. use damp cloths).

- When using cloths and bucket/solution system to clean; Avoid 'double-dipping' used cloths into the bucket containing clean, unused cloths. Doing this can contaminate the remaining clean cloths which are in the solution and result in spreading microorganisms to surfaces that are wiped thereafter

- To maximize the use of cleaning cloths; They should be folded and rotated in a manner so as all surface areas of the cloth, including the front and back, are used progressively as elements are cleaned

Care and Storage of Cleaning Supplies

A. Cleaners' Room:

These rooms should be provided throughout the facility to maintain a clean and sanitary environment. Ideally there should be at least one room per ward dedicated for the cleaner to perform housekeeping duties and should not be used for other purposes. The room should be well ventilated, and the size is determined by the number of rooms to be cleaned and amount of equipment stored. Other standards should be followed as follow:

- Shall be maintained in accordance with good hygiene practices
- Rooms should be inspected on a regular basis to ensure that it is maintained in accordance with good hygiene practices.
- Should have personal protective equipment (PPE) such as eye protection, gloves and gowns/aprons should be available for use. They should be stored in clean cupboards and not open racks where contamination is possible.
- Should have an appropriate water supply for hand washing and a sink/floor drain
- Should have suitable lighting

- Should be easily accessible in relation to the area it serves
- Should be appropriately sized to the amount of materials, equipment, machinery and chemicals stored in the room and allow for proper ergonomic movement within the room.
- Should never contain personal clothing or grooming supplies, food or beverages
- Shall have chemical storage that ensures chemicals are not damaged and may be safely accessed
- Should be free from clutter to facilitate cleaning
- Should be designed so that, whenever possible, buckets can be emptied without lifting them.
- All chemicals disinfectant and materials should be stored above the floor& labeled
- Safety Data Sheet (SDS) should be easily available in case of accidents.

B. Color coded cleaning materials and equipment

The color-coding system helps to ensure that materials and equipment used for cleaning purposes are not used in multiple different areas, therefore reducing the risk of cross-infection. If choosing to implement this system, all cleaning materials and equipment, for example, cloths (re-usable and disposable), mops, buckets, aprons and gloves, should be given a specific color code.

Colour	Designated areas
Red	Operation theaters, delivery suits and Toilets
Blue	General wards, outpatient departments, offices and basins in public areas
Green	Catering department, ward pantry, kitchen areas, patients and staff food service areas
Yellow	Isolation areas, rooms and their toilets

C. Cleaning Equipment:

- Cleaning and disinfection equipment (e.g. mops, buckets, rags/clothes) should be well maintained, in good repair and be cleaned and dried between uses.
- Automated dispensing systems are preferred over manual dilution for accurate calibration.
- Mop heads should be laundered daily and dried thoroughly before storage.
- Cleaning carts should have a clear separation between clean and soiled items, should never contain personal items and should be thoroughly cleaned at the end of the day
- Equipment which generates and disperses dust such as feather dusters and brooms should not be used within the healthcare facility.
- Vacuums used for clinical areas cleaning purpose should be fitted with high-efficiency particulate air (HEPA) filters and undergo regular maintenance, which includes changing the filters on a regular basis (i.e. included in a scheduled maintenance program).
- The use of spray bottles or equipment that might generate aerosols during usage should be avoided. Chemicals in aerosols may cause irritation to eyes and mucous membranes. Containers that dispense liquid such as ‘squeeze bottles’ can be used to apply detergent/disinfectants directly to surfaces or to cleaning cloths with minimal aerosol generation.

- Care needs to be taken to ensure that cleaning cloths are suitable for purpose, that there is sufficient quantity for staff to undertake their duties effectively and that they are used appropriately to prevent cross contamination of surfaces
- When using reusable cleaning cloths, a system should be developed that ensures that a clean cloth(s) is used for each patient area. Failure to do so could compromise the effectiveness of the cleaning process. The cloths should be laundered after each use.
- Disinfectant wipes are useful for decontaminating small items of patient care equipment or high touch surfaces, particularly in clinical outpatient areas, e.g. radiology, ICU, OT and isolation areas/room. They should not be used for cleaning large surface areas since they do not generally contain enough product to cover large areas, and many wipes would be required for effective decontamination

D. Dirty Utility Rooms/ Workrooms:

Each patient/ resident care area should be equipped with a room that may be used to clean dirty patient/ resident equipment that is not sent for central reprocessing (e.g., IV poles, commode chairs). A dirty utility room/ workroom should:

- Be physically separate from other areas, including clean supply/ storage areas
- Be designed to minimize the distance from point-of-care
- Have a work counter and clinical sink (or equivalent flushing-rim fixture).
- Have a dedicated hand washing sink with both hot and cold running water
- Have adequate space to permit the use of equipment required for the disposal of waste
- Have PPE available to protect staff during cleaning and disinfecting procedures
- Be adequately sized within the unit.
- Dirty utility rooms/ workrooms should not be used to store unused equipment.

Cleaning during and after construction/renovation works:

Construction activities generate dust and contaminants that may pose a risk to clients/ patients/ residents, staff or visitors in all health care settings. Cleaning is of importance both during construction and after completion of the construction project.

The level of cleaning should be performed by construction workers to remove gross soil, dust and dirt, construction materials and workplace hazards within the construction zone. This is done at the end of the day, or more frequently if needed, to avoid accumulation of dust. The component which must be done by construction workers inside the construction zone/hoarding:

- Floors are swept to remove debris
- Walk-off mats are vacuumed
- ‘Sticky’ mats are replaced
- Large pieces of drywall, wiring etc. are removed
- Work surfaces may be wiped clean

The following are to be done by hospital cleaning staff in the immediate outside of the construction zone/hoarding:

- Floors and baseboards are free of stains, visible dust, spills and streaks

- Walls, ceilings and doors are free of visible dust, gross soil, streaks, spider webs and handprints
- All horizontal surfaces are free of visible dust or streaks (includes furniture, window ledges, overhead lights, phones, picture frames, carpets etc.)
- Bathroom fixtures including toilets, sinks, tubs and showers are free of streaks, soil, stains and soap scum
- Mirrors and windows are free of dust and streaks
- Dispensers are free of dust, soiling and residue and replaced when empty
- Appliances are free of dust, soiling and stains
- Waste is disposed of appropriately
- Items that are broken, torn, cracked or malfunctioning are replaced

Environmental Cleaning Following Flooding:

In the event of a flood (e.g., overflow from washing machine, dishwasher, toilet, and sewer), the area must be immediately assessed by IP&C (infection control doctor/ICP) to determine the risk of contamination. Until confirmed as a clean water source, all staff should assume that the water is contaminated.

Immediate contamination may occur if the source of the flood water harbors pathogenic bacteria (e.g., sewer or toilet overflow) and the area will need to be cordoned off until cleaning and disinfection are completed.

If the flooding involves a food preparation area, all food products that have come into contact with flood water must be discarded and Public Health notified. Public Health must also be notified if vaccine refrigerators are involved in a flood or if flooding leads to a prolonged power outage that compromises food or vaccine refrigeration.

Steps to Take for Infection Prevention and Control in the Event of a Flood (sample procedure)

- Assess patient, visitor and staff safety evacuate the area if required
- Protect affected equipment with plastic sheeting or move if possible
- Contain the flood if possible
- Disinfect surfaces of equipment and furniture before moving it from the flood area using hospital approved detergent and disinfectant
- Notify Infection Prevention and Control to assess the risk of contamination
- If water is contaminated with faecal material, ICP should be consulted regarding staff and patient safety.
- ICP will arrange for ongoing patient surveillance dependent on the patient population affected by the flood
- ICP will recommend relocation of patients if required dependent on patient population.
- Following containment:
    - discard all contaminated single-use sterile supplies
    - send contaminated reusable sterile supplies to be reprocessed
    - remove and discard contaminated carpeting
    - assess furniture and equipment to determine if it can be salvaged
    - assess building materials (e.g., ceiling tiles, drywall) and remove if required
    - Clean and disinfect the area affected

Education & Training

- Environmental cleaning is a team effort. Personnel responsible for cleaning the environment and equipment must receive education and training on proper environmental cleaning and disinfection methods, agent use and selection, and safety precautions.
- All housekeepers must undergo a documented orientation session.
- The training should be completed before new staff members are allowed to work without direct supervision.
- The orientation program should be based on standard operating procedure (SOP) include cleaning techniques, highlighting on the high touch areas, cleaning agents and infection control practices e.g.: blood borne pathogens, isolation precautions and standard precautions, N95 particulate respirator and waste disposal. As a minimum, training must be given in the performance of cleaning tasks, the use of cleaning equipment, control of infection, employee health, manual handling, fire, health and safety and site orientation.
- Training should be repeated in its entirety every year or sooner if a competency issue has been identified. Training should always be delivered by a suitably qualified person.
- Staff should be assessed on their competency after the training to evaluate their level of understanding. The training should be repeated in its entirety every year for the housekeepers or sooner if a competency issue has been identified. All housekeepers are recommended to have written training records that are signed and dated by the trainer and trainee. The training record should be stored safely in each employee's personal file.

Assessment of Cleanliness and Quality Control

- Housekeeping is responsible to ensure that the quality of cleaning maintained in the hospital and meets appropriate infection prevention and control best practices.
- Documentation of cleaning timing and dates with signature of staff did it and counter checked and signed by the cleaning supervisor must be maintained and a form must be kept visible in the area.
- The responsibility for ensuring that the standardized cleaning practices are adhered to lies not just with the person performing the task but also with the direct supervisor and management of the department. It is important to incorporate elements of quality improvement into an enhanced cleaning program, including monitoring, audits and feedback to staff and management.
- The infection prevention and control department conduct minimum of one detailed environmental audit for all areas in the hospital. Feedback to be given to the respective department with involvement of housekeeping staff.
- The effectiveness and quality of cleaning is monitored by the following:
    - **Visual inspection of cleanliness using checklist:** It is recommended to be done by multidisciplinary team.
    - **Visual assessment of cleaning technique:** supervisors review cleaning techniques of the cleaning staff to ensure they are in accordance with documented policies and procedures.

- **Fluorescent markers:** high touch surfaces are marked with a fluorescent marker before cleaning occurs. A "black" light (i.e. ultraviolet (UV)) is then used after cleaning has been completed to assess whether the marker has been removed and hence the surface adequately cleaned.
- **ATP testing methods**: these are commercial systems designed to test for residual organic matter on surfaces, and are commonly used in the food industry.
- **Microbiological tests:** routine microbiological sampling of the environment to determine the effectiveness of cleaning has considerable limitations and is not generally recommended. However, microbiological testing can be used as part of outbreak management investigations as per infection prevention and control team decision.

- Monitoring should be an ongoing activity built into the routine cleaning regimen.
- Periodical monitoring should take place immediately after cleaning to ensure that the cleaning has been carried out to an appropriate and agreed standard. Data from monitoring should be retained and used in trend analysis and compared with benchmark values that have been obtained during the validation of the cleaning program.

Appendix A: **Components of basic routine cleaning applicable to administrative, Doctors and other staff offices, engineering workshop, plant rooms and cafeteria.**

Box1: Components of basic routine cleaning

- Floors and baseboards are free of stains, visible dust, spills and streaks
- Walls, ceilings and doors are free of visible dust, gross soil, streaks, spider webs and handprints
- All horizontal surfaces are free of visible dust or streaks (includes furniture, window ledges, overhead lights, phones, picture frames, carpets etc.)
- Bathroom fixtures including toilets, sinks, tubs and showers are free of streaks, soil, stains and soap scum
- Mirrors and windows are free of dust and streaks
- Dispensers are free of dust, soiling and residue and replaced/replenished when empty
- Appliances are free of dust, soiling and stains
- Waste is disposed of appropriately
- Items that are broken, torn, cracked or malfunctioning are replaced
- Curtains are free from visible dirt, damaged or torn are replaced.

Appendix B: Risk Stratification Matrix to Determine Frequency of Cleaning:

FOR EACH CLIENT/ PATIENT/ RESIDENT AREA or DEPARTMENT:

STEP 1: Categorize the factors that will impact on environmental cleaning:

Probability of contamination with pathogens

Heavy Contamination (score = 3)
An area is designated as being heavily contaminated if surfaces and/or equipment are routinely exposed to copious amounts of fresh blood or other body fluids (e.g., birthing suite, autopsy suite, cardiac catheterization laboratory, haemodialysis station, Emergency room, client/patient/resident bathroom if visibly soiled).

Moderate Contamination (score = 2)
An area is designated as being moderately contaminated if surfaces and/or equipment does not routinely (but may) become contaminated with blood or other body fluids and the contaminated substances are contained or removed (e.g. wet sheets). All client/patient/resident rooms and bathrooms should be considered to be, at a minimum, moderately contaminated.

Vulnerability of population to environmental infection

More Susceptible (Score = 1)
Susceptible clients/patients are those who are most susceptible to infection due to their medical conditions or due lack of immunity (oncology, transplant, neonates and those with severe burns.

Less Susceptible (Score = 0)
All other individual and areas are classified less susceptible.

Potential for exposure

High touch surfaces (Score 3):
High touch surfaces are those that have frequent contact with hands. Examples include doorknobs, telephone, call bells, bedrails, light switches wall areas around the toilet & edges of privacy curtains.

Low touch surfaces (Score 1):
Those that have minimal contact with hands examples include walls, ceilings, mirrors & windows

STEP 2: Determine the total risk stratification score:
For each functional area or department, the frequency of cleaning is based on the factors listed in the boxes above. A score is given if the factors are present, and the frequency of cleaning is based on the total score as derived in the following matrix:

Probability of contamination	Potential for exposure				
	High-Touch Surfaces (Score 3)			Low-Touch Surfaces (Score =1)	
	More Susceptible Population (Score 1)	Less Susceptible Population (Score = 0)			
Heavy (Score = 3)	7	6		5	4
Moderate (Score = 2)	6	5		4	3
Light (Score =1)	5	4		3	2

STEP 3: Determine the cleaning frequency based on the risk stratification matrix:
Depending on the total score resulting from the risk stratification matrix above, cleaning frequencies for each functional area or department are derived:

Total Risk Score	Risk Type	Minimum Cleaning Frequency
7	High Risk	Clean after each case/event/procedure and at least twice per day Clean additionally as required
4-6	Moderate Risk	Clean at least once daily Clean additionally as required (e.g., gross soiling)
2-3	Low Risk	Clean according to a fixed schedule Clean additionally as required (e.g., gross soiling)

Total risk score for different hospital areas based on risk stratifications matrix:

Location	**Probability of contamination**	**Potential for exposure**	**Population**	**Total Score**
	Light = 1 Moderate = 2 Heavy = 3	High-Touch = 3 Low-Touch = 1	Less Susceptible = 0 More Susceptible = 1	
Operating Theatres	3	3	1	7
AICUs, PICUs, NICU, High dependency area	3	3	1	7
Delivery Suite	3	3	1	7
Cardiac Catheterization lab, interventional radiology,	3	3	1	7

Cystoscopy	3	3	0/1	6/7
oncology ward, hematology ward, transplant area, chemotherapy unit	3	3	1	7
Dental Procedure room	3	3	0/1	6/7
Emergency department	3	3	0/1	6/7
Dialysis unit	3	3	1	7
Labour room	3	3	0/1	6/7
CSSD	3	3	0	6
Laboratory	3	3	0	6
General pediatric, surgical, medical wards,	3	3	0	6
outpatient department	2	0	3	5
Laundry soiled linen area	3	3	0	6
Physiotherapy	2	1	0/1	3/4
Pharmacy	1	0	3	4
mortuary	1	1	0	
Offices	1	1	0	2
Engineering workshops, plant room, education department	1	1	0	2

Appendix C: Cleaning frequency and procedure summary for the hospital areas

Area Description	Frequency	Products/Technique	Additional Guidance / Description of Cleaning
Adult OPD Waiting/ admission areas (Adult)	At least daily and as needed	Clean (neutral detergent and water): • high-touch surfaces • floors	In addition, clean low-touch surfaces on a scheduled basis (e.g., weekly)
Adult OPD Consultation/ examination areas (Adult)	At least twice per day and as needed	Clean (neutral detergent and water): • high-touch surfaces	Last clean of the day: clean the entire floor with neutral detergent and water In addition, clean low-touch surfaces on a scheduled basis (e.g., weekly)
Pediatric OPD Waiting / admission areas	At least daily and as needed (e.g., visibly soiled, blood/body fluid spills)	Clean and disinfect: • high-touch and low-touch surfaces • floors	Toys that may be put into mouth of infant or toddler must be cleaned, disinfected and rinsed thoroughly after each use. Other toys that are not in contact with body fluids can be disinfected with wipes between patients.
Pediatric OPD Consultation / examination areas	After each event/ case and at least twice per day and as needed	Clean and disinfect: • high-touch surfaces	Last clean of the day: clean and disinfect the entire floor and low- touch surfaces
OPD Minor operative procedure rooms (Adult or Pediatric)	Before and after (i.e., between) every procedure	Clean and disinfect: • any surface visibly soiled with blood or body fluids • high-touch surfaces in the patient zone • floors in the patient zone	Last clean of the day: clean and disinfect: • other high-touch surfaces and low-touch surfaces • hand washing sinks • scrub/sluice areas • the entire floor
ER Waiting / admission areas	At least daily and as needed (e.g., visibly soiled, body fluid spills)	Clean and disinfect: • high-touch and low-touch surfaces • floors	In addition, clean low-touch surfaces on a scheduled basis (e.g., weekly). Rinse with water at the end of deep cleaning.
ER Consultation/ examination areas	After each event/ case and at least twice per day and as needed	Clean and disinfect: • high-touch surfaces	Last clean of the day: clean and disinfect the entire floor and low- touch surfaces

ER Procedure areas include trauma areas for high acuity patients	Before and after (i.e., between) every procedure	Clean and disinfect: • any surface visibly soiled with blood or body fluids • high-touch surfaces in the patient zone • floors in the patient zone	Last clean of the day: clean and disinfect: • other high-touch surfaces and low-touch surfaces • handwashing sinks • scrub/sluice areas • the entire floor
Toilets for general inpatient and outpatient areas; frequently used by visitors, family members	At least once daily (private patient room) Public and shared toilets are cleaned more frequently as needed	Clean and disinfect: • high-touch/frequently contaminated surfaces handwashing sinks faucets handles toilet seat door handles • floors • any surface visibly soiled with blood or body fluids	In addition, clean low-touch surfaces on a scheduled basis (e.g., weekly). Rinse with water at the end of deep cleaning.
Floors in general inpatient and outpatient areas, always cleaned last after other environmental surfaces	At least daily	Clean (neutral detergent and water): • clean to dirty, systematic manner (figure-eight pattern, regularly rinse in rinse bucket)	Floors may require, depending on the risk-level in a specific patient care area: • more frequent cleaning • use of a disinfectant
Any spill in any patient or non- patient area	Immediately, as soon as possible	1. Wear appropriate PPE 2. Confine the spill and wipe it up immediately with absorbent (paper) towels, cloths, or absorbent granules (if available) that are spread over the spill to solidify the blood or body fluid (all should then be disposed as infectious waste). 3. Clean (neutral detergent and water). 4. Disinfect using 10,000ppm chlorine based disinfectant 5. Immediately reprocess all reusable supplies and equipment (e.g., cleaning cloths, mops) after the spill is cleaned up.	Mark off spill area to prevent contact, as well as accidental slips and falls

Operating room	Before first procedure	Disinfect: • horizontal surfaces furniture surgical lights operating bed stationary equipment	See Operating rooms Records of previous evening terminal clean required; if not or if no surgeries on the day prior, perform terminal clean (as below)
	Before and after every procedure	Clean and disinfect: • high-touch surfaces (e.g., light switches, door knobs) outside surgical field • any surface visibly soiled with blood or body fluids • all surfaces and noncritical equipment and the floor inside the surgical field	Remove all used linen and surgical drapes, waste (including used suction canisters, ¾ filled sharps containers), and kick buckets, for reprocessing or disposal Portable noncritical (e.g., compressed gas tanks, x-ray machine) equipment should be thoroughly cleaned and disinfected before and after each procedure
	After last procedure (terminal clean)	Clean and disinfect: • all surfaces and noncritical equipment in the operating room • the entire floor • any surface visibly soiled with blood or body fluids • scrub and utility areas/sinks	Take care to move the operating table and any mobile equipment to make sure that the floor areas underneath are thoroughly cleaned and disinfected Clean and disinfect low-touch surfaces, (e.g., the insides of cupboards and ceilings/walls) on a scheduled basis (e.g., weekly)
Medication preparation areas	Between uses	Clean and disinfect: • countertops • portable carts used to transport or prepare medications	
	End of each day	Clean and disinfect: • high-touch surfaces • floors	Clean and disinfect low-touch surfaces, such as the tops of shelves and walls/ vents, on a scheduled basis (e.g., weekly)
Reprocessing areas	Before and after every use	Clean and disinfect: • utility sinks used for washing • semi-critical equipment (e.g., endoscopes)	
	Twice daily	Clean and disinfect: • all high-touch surfaces countertops surfaces of washing equipment handwashing sinks	

Critical units	Twice daily and as needed	Clean and disinfect: • high-touch surfaces (only outside of neonatal incubator when occupied) Clean: • floors with neutral detergent and water	Last clean of the day: clean low-touch surfaces
	At discharge/ transfer (terminal cleaning)	Clean and disinfect: • high-touch surfaces • low-touch surfaces • floors	Follow terminal cleaning procedure
Isolation areas except diarrhea and Candida auris and diarrhea	Routine cleaning	Daily Clean and disinfect: • high-touch surfaces, with focus on the patient zone • floors	In addition, clean low-touch surfaces on a scheduled basis (e.g., weekly) Cleaning staff must wear required PPE Table 5 (page 36) Dispose of or reprocess cleaning supplies and equipment immediately after cleaning Last clean of the day: clean and disinfect the entire floor and low- touch surfaces
Isolation areas occupied by patients with Candida auris or diarrhea (including C. difficile)	Routine cleaning	Daily Clean and disinfect (two- step process required and sporicidal agent): • any surface visibly soiled with blood or body fluids • high-touch surfaces in the patient zone • floors	
All isolation areas	At discharge/ transfer (terminal cleaning)	Clean and disinfect: • high-touch surfaces • low-touch surfaces • floors	Follow terminal cleaning procedure. For Diarrhea and Candida auris, two- step process required and sporicidal agent(eg: chlorine based disinfectant)

General procedure areas (such as radiology and endoscopy services)	Before and after every procedure (i.e. between)	Clean and disinfect:
		• any surface that is visibly soiled with blood or body fluids
		• high-touch surfaces inside the patient zone
		    procedure table/station
		    counter tops
		    external surfaces of fixed equipment
		• floors inside the patient zone
			Remove disposable equipment and reprocess reusable noncritical patient care equipment

	After last patient of the day (terminal clean)	Clean and disinfect: • all high-touch and low-touch surfaces • entire floor	Move the procedure table and other portable equipment to clean and disinfect the entire floor area Handwashing sinks should be thoroughly cleaned (scrubbed) and disinfected
Delivery rooms	Before and after (i.e., between) every procedure	Clean and disinfect: • any surface that is visibly soiled with blood or body fluids • high-touch surfaces inside the patient zone • floor inside the patient zone	Remove soiled linens and waste containers for disposal/reprocessing
	After last delivery of the day (terminal clean)	Clean and disinfect: • any surface that is visibly soiled with blood or body fluids • all high-touch and low-touch surfaces • entire floor	Move the procedure table and other portable equipment to clean and disinfect the entire floor area Handwashing sinks should be thoroughly cleaned (scrubbed) and disinfected

Hemodialysis stations/areas	Before and after every procedure (i.e. between)	Clean and disinfect:
		• any surface that is visibly soiled with blood or body fluids
		• all surfaces of the dialysis station area
		  bed
		  chair
		  countertops
		  external surfaces of the machine
		• floor inside the patient zone
			Remove disposable patient care items/ waste and reprocess reusable patient care equipment per below
			Take care to allow enough contact time before the next subsequent use of the station/area

	After last patient of the day (terminal clean)	Clean and disinfect: • any surface that is visibly soiled with blood or body fluids • all surfaces of the dialysis station/area • high-touch surfaces in the area • entire floor	Move the procedure table and other portable equipment to clean and disinfect the entire floor area In addition, clean low-touch surfaces on a scheduled basis (e.g., weekly)

References

Title of book/ journal/ articles/ Website	Author	Year of publication
Cleaning standard for South Australia healthcare facilities	Government of South Australia	2017
Infection Prevention and Control Cleaning Policy	East Cheshire, NHS trust, UK	2016
Cleaning Standards for Healthcare Facilities	Government of South Australia	2014
Best Practices for Environmental Cleaning for Prevention and Control of Infections In All Health Care Settings - 2nd edition	Provincial Infectious Diseases Advisory Committee,	2012
Best practices for Environmental cleaning in healthcare facilities	CDC	2019

	Revision History		
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1	**Initial Release**	Zaina AL Maskari	01.6.2017
2	**Reviewed by**	Dr. Zaina Al Maskari	15.9.2019
3	**Reviewed by**	Aisha Al Mamary	10.10.2022
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Written by	Checked by	Authorized by
Zaina AL Maskari	Dr. Omaira Al Omairi	Dr. Amal Al Tai